Bridge Day 1 STUDENT WORKBOOK

Visible Output and Stored State

Learning sequence: Predict - Trace - Explain - Test - Interpret

Guided engineering evidence workbook

Name		**UIN**		**Section**	

Concrete engineering situation

Your team is doing a first-day hallway cart check. A small wheeled cart is rolled along a taped 12-meter path. One teammate times the run with a stopwatch. Another teammate enters the values in a Python notebook so the team can decide what should go into the lab record.

The problem today is not advanced physics. The problem is evidence: what did the notebook store, what did it visibly show, and which value is safe to write in the record?

Engineering task	Build a reliable evidence trail for two hallway cart trials before a result is copied into a lab record.
Computational move	Separate measurement inputs, stored variables, printed output, notebook display, and later overwrites.
Python focus	assignment, arithmetic expression, variable overwrite, `print`, final-expression notebook display, state after each cell.
Evidence to keep	exact visible lines, displayed values, stored values after each cell, a corrected lab note, and one non-mutating check.

Day 1 evidence rules

1. Assignment stores or overwrites a value. Example: `speed_mps = distance_m / time_s`.
2. `print(...)` creates visible output but does not by itself change a variable.
3. In a notebook, a bare expression at the end of a cell can display a value. Displaying a value is not the same as storing it.
4. A later assignment can overwrite an earlier stored value. Your trace must say what is stored *after each cell*, not just what looked important on the screen.

1. Read the notebook as an evidence system

recognize predict

Use the scenario above to identify what kind of evidence each line can create. Do not calculate everything yet.

```
# Cell 1: first run on the hallway path
trial_id = "cart-A"
distance_m = 12.0
time_s = 3.0
speed_mps = distance_m / time_s
print("trial", trial_id)
print("speed", speed_mps)

# Cell 2: second run, same path, new time
trial_id = "cart-B"
time_s = 2.4
speed_mps = distance_m / time_s
print("trial", trial_id)
speed_mps + 0.2

# Cell 3: keep the adjustment separate
adjusted_speed = speed_mps + 0.2
print("stored", speed_mps)
print("adjusted", adjusted_speed)

# Cell 4: decide to record the adjusted value
speed_mps = adjusted_speed
print("recorded", speed_mps)
```

Pts.	Prompt	My evidence
1	Which variable names store measurement inputs rather than calculated results?	
1	Which line first calculates a speed from the measured distance and time?	
1	Which line in Cell 2 displays a candidate adjusted value without storing that adjustment?	
1	Which line in Cell 4 overwrites the stored value of `speed_mps`?	

2. Trace stored state after each cell

trace predict

Complete the state table. If a variable does not exist yet, write `not defined`. The first row has hints; the later rows require you to track overwrites.

Checkpoint	`trial_id`	`time_s`	`speed_mps`	`adjusted_speed`	Reason for any change
After Cell 1	`cart-A`	`3.0`		`not defined`	
After Cell 2					
After Cell 3					
After Cell 4					

Pts.	Prompt	My response
2	At what checkpoint does `speed_mps` first become `5.0`? Explain the measurement change that caused it.	
2	At what checkpoint does `speed_mps` first become `5.2`? Explain what line caused the overwrite.	

3. Separate printed output from notebook display

predict trace

Visible evidence can come from printed output or from a final notebook expression. Complete both columns. If a cell has no notebook display, write `none`.

Cell	Printed output lines from `print(...)`	Notebook display from final bare expression
Cell 1		
Cell 2		
Cell 3		
Cell 4		

Common trap to check

A displayed value can look official because it appears on the screen. But a lab record should say whether the value was printed, displayed as a candidate calculation, or stored as the current value after an assignment.

Pts.	Prompt	My response
2	In Cell 2, why is the displayed `5.2` not yet the stored value of `speed_mps`?	
2	In Cell 3, why is it useful to print both `stored` and `adjusted`?	

4. Debug the lab record

debug explain test

A teammate writes this lab note after Cell 2:

Lab note to debug

Trial cart-B speed was recorded as 5.2 m/s because the notebook showed 5.2.

Pts.	Prompt	My response
2	What is accurate in the teammate's note?	
2	What is inaccurate or incomplete in the teammate's note?	
2	Rewrite the note so it separates measured speed, displayed candidate adjustment, and the value actually recorded after Cell 4.	

Pts.	Prompt	My response
1	Write one line you could run after Cell 2 to check the stored value of `speed_mps` without changing it.	
1	Write one line you could run after Cell 3 to check whether `adjusted_speed` exists.	
2	Before calculating speed, what measurement condition should the team check so the result is physically meaningful? Write it in plain language.	

5. Compare two possible edits

debug implement explain

The team considers two ways to apply the 0.2 m/s adjustment. Both can show `5.2`; they do not leave the same evidence trail.

Version A: keep candidate separate

```
adjusted_speed = speed_mps + 0.2
print("stored", speed_mps)
print("adjusted", adjusted_speed)
```

Version B: overwrite immediately

```
speed_mps = speed_mps + 0.2
print("stored", speed_mps)
print("adjusted", speed_mps)
```

Pts.	Prompt	My response
1	In Version A, what is stored in `speed_mps` after the code runs?	
1	In Version A, what is stored in `adjusted_speed` after the code runs?	
1	In Version B, what is stored in `speed_mps` after the code runs?	
2	Which version gives a better audit trail for a lab record? Explain your choice.	
2	If the goal is to record the adjusted value only after review, write one sentence explaining why Cell 4 is a safer place to overwrite `speed_mps`.	

6. Mastery check and support next step

transfer explain

Use your own evidence from the workbooklet. Do not write a generic answer.

Pts.	Prompt	My response
2	What is one example from this notebook where the screen showed a value that was not yet the stored value that would be recorded?	
2	In one or two sentences, explain why engineers should preserve both visible output and stored-state evidence during early computational checks.	

My mastery check

Mark the strongest statement that is true after this workbooklet.

☐	Secure: I can trace stored state and visible output separately, including a displayed value that is not stored yet.
☐	Developing: I can calculate some values, but I still need support separating display, print, assignment, and overwrite.
☐	Needs support: I need a smaller example before I can explain what Python stored after each cell.

My next support move

My issue is mostly: setup / Python syntax / tracing / arithmetic / notebook display / confidence / time.
The first place I got uncertain was: _____
My next small action is: ask a partner / ask instructor / rerun a cell / rebuild the trace table / try the recovery version.

Workbooklet target: I can build an evidence trail that separates measurement inputs, stored variable state, printed output, notebook display, and a later overwrite before a result enters an engineering record.

Bridge Day 1 STUDENT WORKBOOK

VIP Evidence Handoff

Learning sequence: Predict - Trace - Explain - Test - Interpret

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Why engineers use this page

Carry exact-output and stored-value evidence directly into the first notebook, then complete its two-problem Independent Program Studio.

Decision boundary: Code is useful only when its stored state, visible result, assumptions, units, limits, and tests support a defensible engineering interpretation.

Construct readiness before independent work

New authoring tools: string literal, assignment, print call, sequential execution, exact output
Carried authoring tools: none
Supplied-code exposure only: none
An exposure-only construct may be traced in complete supplied code, but the independent task will not require students to invent it before its authoring day.

Notebook cells and task constructs

Activity	Work	Stable cells	Required authoring constructs
18.1	First Output	`18-1-work / 18-1-transfer / 18-1-cold`	profile-level contract
18.2	Formatted Notice	`18-2-work / 18-2-transfer / 18-2-cold`	profile-level contract

Readiness evidence

1. For Activity 18.1, separate the exact fixed label from the value stored in the named variable.

2. For Activity 18.2, number the required output lines and write the exact order before opening the notebook.

Timing and help trigger

Record a first prediction before running. If you still lack an executable plan after about 8 focused minutes, use the linked ThinkCSPy refresher or the supplied model. If two attempts fail for the same named requirement, write the exact mismatch and ask a peer mentor or instructor a targeted question. Timing is diagnostic evidence, not a speed grade. **Start or elapsed minutes:** _____ **My help trigger:** _____

Engineering Evidence Record

Complete one record from a model, transfer, or Independent Program Studio build. Generic statements are not sufficient; use actual values, a real revision, and one concrete next test.

Evidence field	Record
Prediction	
Observed Result	
Revision Reason	
Engineering Meaning	
Units Or Not Applicable	
Assumption	
Model Limit	
Next Test	
Help Trigger	
Minutes Used	

Notebook and submission procedure

1. Attempt, predict, run, inspect, and revise before opening a reveal.
2. Complete the end-of-notebook Independent Program Studio without a reveal or copied solution. Only those visibly labeled Studio cells are scored for independent mastery.
3. Save or download the completed notebook as one `.ipynb` file.
4. Upload that one notebook to the matching Gradescope assignment. Do not export a `.py` file or create a ZIP.

Generated from the canonical VIP construct and evidence contract; workbook planning and executable notebook work remain distinct.